Circuit tools and semiconductor physics

Practice questions · Day 01 · Sat 24 Oct · pack E1

5. Practice

From your course files

WS1 Q1 (course question). When there are more *electrons* than *protons* in a piece of wire, we say it has a ———— charge.

a) Negative
b) Positive
c) Neutral
d) Not enough information to answer

WS1 Q2 (course question). In Figure 1, i_1 is positive if positive charges are moving to the ———— or *negative* charges are moving to the ————.

a) Left, Right
b) Right, Right
c) Left, Left
d) Right, Left



WS1 Figure 1: conventional current direction (from the workshop sheet)

WS1 Q3 (course question). If our assumption about the higher potential end is wrong, the potential difference (voltage) will turn out to be ————.

a) Positive
b) Zero
c) Negative
d) Not enough information to answer.

WS1 Q4 (course question). Note the Passive sign convention for Ohm's law, Current flows from the higher potential to the lower potential end. Therefore, charge carriers ———— potential energy by moving from the ———— end of the voltage to the ———— end. The energy is ———— by the resistor.

a) Lose, Positive, Negative, Absorbed
b) Lose, Negative, Positive, Absorbed
c) Absorb, Positive, Negative, Released
d) Absorb, Negative, Positive, Released

WS1 Q5 (course question). In a resistor with current $i(t)$, the power is ————.

a) Always negative
b) Always positive
c) Negative or positive
d) Negative or positive or zero

WS1 Q6 (course question). In a capacitor, if the voltage is constant, current is zero, and the capacitor is ———— to DC.

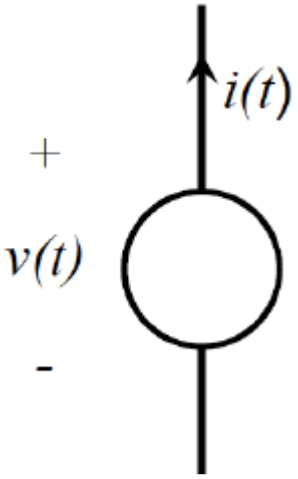
a) Short circuit
b) Open circuit
c) Constant voltage source
d) Either open or short circuit depending on the voltage magnitude

WS1 Q7 (course question). In an inductor, if the current is constant, voltage is zero, and the inductor is ———— to DC.

a) Short circuit
b) Open circuit
c) Constant current source
d) Either open or short circuit depending on the voltage magnitude

WS1 Q8 (course question). In the voltage source shown in Figure 5, current is assumed to flow from ———— potential end to the ———— potential end. If the source is supplying energy, $i(t)$ will be ————.

a) Low, High, Positive
b) High, Low, Negative
c) Low, High, Negative
d) High, Low, Positive

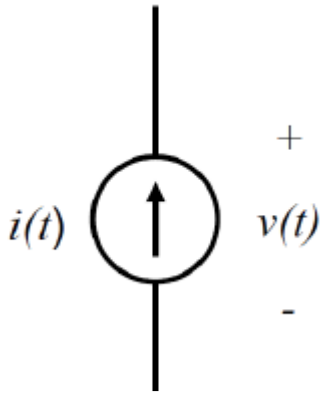


WS1 Figure 5: independent voltage source symbol, active notation (from the workshop sheet)

WS1 Q9 (course question). In the current source shown in Figure 6, if the source is supplying energy, $v(t)$ will be ————. Positive current leaves ———— end of the voltage.

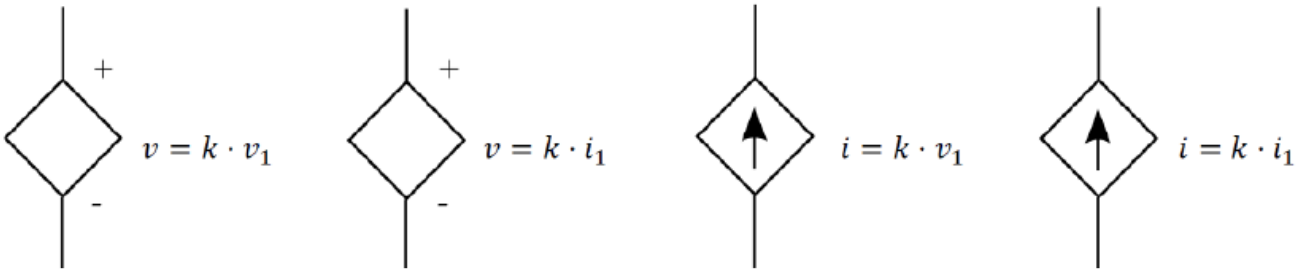
a) Positive, Negative
b) Positive, Positive

c) Negative, Positive
d) Negative, Negative



WS1 Figure 6: independent current source symbol, active notation (from the workshop sheet)

WS1 Q10 (course question). Identify the type of dependent/controlled sources from left to right in Figure 7.



WS1 Figure 7: dependent/controlled source symbols (from the workshop sheet)

WS1 Q11 (course question). What does short circuit mean? Select all correct options.
a) Zero potential difference
b) zero current
c) 0 Ω resistance
d) 0 S conductance
e) infinite resistance
f) infinite conductance

WS1 Q12 (course question). Which statement(s) is(are) correct about Passive Elements? Select all correct options.

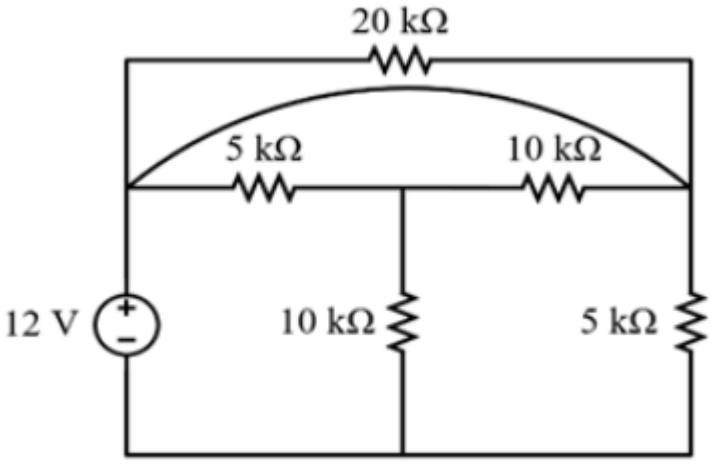
a) Deliver power when $p(t) = v(t) \times i(t) < 0$

b) Deliver power when $p(t) = v(t) \times i(t) > 0$

c) Absorb power when $p(t) = v(t) \times i(t) > 0$

d) Absorb power when $p(t) = v(t) \times i(t) < 0$

WS1 Q13 (course question). How many nodes and branches are there in the circuit given below:



WS1 Q13, Figure 9 (from the workshop sheet)

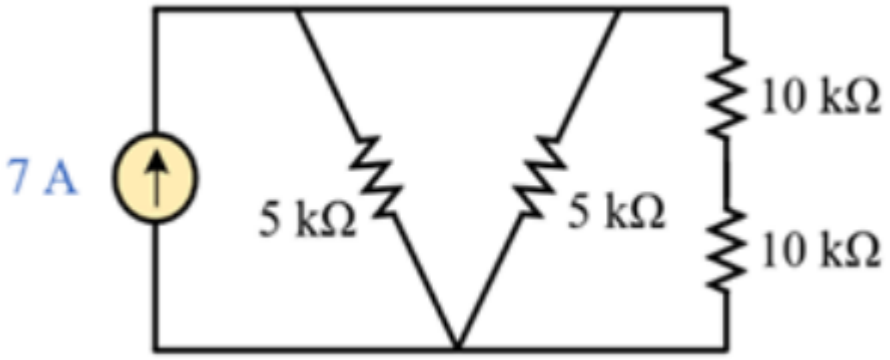
a) $N = 4,\ B = 6$

b) $N = 4,\ B = 7$

c) $N = 3,\ B = 5$

d) $N = 3,\ B = 7$

WS1 Q14 (course question). How many loops and meshes are there in the circuit given below:



WS1 Q14, Figure 10 (from the workshop sheet)

a) Loop=6, Mesh=3
b) Loop=3, Mesh=6
c) Loop=6, Mesh=6
d) Loop=6, Mesh=3

WS1 Q15 (course question). Determine currents I_1 and I_2 in the circuit shown in Figure 11.

WS1 Q15, Figure 11 (from the workshop sheet)

a) $I_1 = 0$ A, and $I_2 = 6$ A
b) $I_1 = 6$ A, and $I_2 = 3$ A
c) $I_1 = 0$ A, and $I_2 = 3$ A
d) $I_1 = 3$ A, and $I_2 = 3$ A

WS1 Q16 (course question). Which option shows the KVL equation for the loop shown in the circuit given in Figure 12?

WS1 Q16, Figure 12 (from the workshop sheet)

a) $-v_1 + v_4 - v_3 + v_2 = 0$
b) $-v_1 + v_4 - v_3 + v_2 + v_5 = 0$
c) $-v_1 + v_4 - v_3 + v_2 - v_5 = 0$
d) $v_1 + v_4 + v_3 + v_2 + v_5 = 0$

WS1 Q17 (course question). What is the equivalent voltage source in the circuit given in Figure 15? Consider the polarity of the 9 V source as the reference for the equivalent source

WS1 Q17, Figure 15 (from the workshop sheet)

a) 6 V
b) −16 V
c) 16 V
d) −6 V

WS1 Q18 (course question). In the circuit shown in Figure 18, what is the value of current I_1?

WS1 Q18–Q19, Figure 18 (from the workshop sheet)

a) 1.2 A
b) 4.8 A
c) 3.6 A
d) 4.5 A

WS1 Q19 (course question). In the circuit shown in Figure 18, what is the voltage value V_2?
a) 9.6 V

b) 2.4 V
c) 7.2 V
d) 9 V

WS1 Q20 (course question). Using Superposition technique, calculate I_x in the circuit shown in Figure 19.

WS1 Q20, Figure 19 (from the workshop sheet)

a) 0.8 A
b) 0.96 A
c) 0.66 A
d) 0.14 A

WS1 Q21 (course question). What is the Thevenin equivalent of the circuit seen from the AB terminals in the circuit shown in Figure 21?

WS1 Q21–Q22, Figure 21 (from the workshop sheet)

a) $V_{AB} = 45$ V, and $R_{AB} = 3\ \Omega$
b) $V_{AB} = 27$ V, and $R_{AB} = 3\ \Omega$
c) $V_{AB} = 27$ V, and $R_{AB} = 9\ \Omega$
d) $V_{AB} = 45$ V, and $R_{AB} = 9\ \Omega$

WS1 Q22 (course question). What is the Norton equivalent of the circuit seen from the AB terminals in the circuit shown in Figure 21?

a) $I_{AB} = 12$ A, and $R_{AB} = 9\ \Omega$
b) $I_{AB} = -15$ A, and $R_{AB} = 9\ \Omega$
c) $I_{AB} = -12$ A, and $R_{AB} = 3\ \Omega$
d) $I_{AB} = 15$ A, and $R_{AB} = 3\ \Omega$

WS1 Q23 (course question). Using source transformation, find the current I_x in the circuit shown in Figure 23?

WS1 Q23, Figure 23 (from the workshop sheet)

a) $I_x = 0.2319$ mA
b) $I_x = 0.1923$ mA
c) $I_x = 0.3129$ mA
d) $I_x = 0.9321$ mA

WS1 Q24 (course question). Using Nodal Analysis, determine voltage V_c in the circuit shown in Figure 24?

WS1 Q24–Q25, Figure 24 (from the workshop sheet; values are conductances in ℧ = S)

a) $V_c = 0.3556$ V
b) $V_c = 0.244$ V
c) $V_c = 0.2$ V
d) $V_c = 0.1$ V

WS1 Q25 (course question). Using Mesh Analysis, determine voltage V_{AB} in the circuit shown in Figure 24?
a) $V_c = 0.3556$ V
b) $V_c = 0.244$ V
c) $V_c = 0.2$ V
d) $V_c = 0.1$ V

WS2 Q1 (course question). Answer the following true or false questions by discussing them in your group: (please note that IX and X are related to diode):
I. At absolute zero temperature, the semiconductor behaves as a conductor.
II. No free electron or hole exists at absolute zero temperature.
III. At room temperature with thermal agitation, we have enough electrons and holes in a silicon wafer to have conductance.
IV. Holes are the majority charge carriers in a p-type semiconductor.
V. A hole is defined as the absence of an electron.
VI. A pure silicon has four valence electrons.
VII. A doped semiconductor is electrically charged at room temperature.
VIII. Diffusion current is the electric current caused by the charge carriers when they move from a higher concentration region to a lower concentration region.
IX. In an unbiased PN junction, the drift/saturation and diffusion currents are equal.
X. In the boundary of the two adjacent layers in a PN junction, we have the depletion layer and barrier voltage.

WS2 Q2 (course question). The cylinder shown here in Fig. 1 is made of copper, whose conductivity is $\sigma = 5.65 \times 10^5\ (\Omega\text{-cm})^{-1}$. Calculate the resistance of the cylinder when it conducts a current lengthwise.

WS2 Fig. 1: the cylinder, 8 mm long and 1 mm in diameter (from the workshop sheet)

WS2 Q3 (course question). The cylinder is now made of intrinsic silicon,
(a) Calculate its resistance at room temperature
(b) Compute the average drift velocities of holes and electrons when 5 volts is applied to this resistor. At room T, $n_i = 1.5 \times 10^{10}$ cm⁻³, $\mu_n = 1350$ cm²/Vs, $\mu_p = 500$ cm²/Vs.

WS2 Q4 (course question). The intrinsic concentration, n_i, of an n-type semiconductor is 1.5×10^{10} cm⁻³ at room temperature.
(a) Estimate the concentrations of free electrons and holes in this material at room temperature if $N_D = 10^{14}$ cm⁻³.
(b) Find the room temperature conductivity of the doped material if the electron and hole mobilities are 1350 and 500 cm²/Vs, respectively.

WS2 Q5 (course question). The intrinsic concentration of acceptor-type silicon has values of 1.5×10^{10} cm⁻³ and 10^{13} cm⁻³ at 300 K and 405 K, respectively, and the doping concentration is 2×10^{15} cm⁻³.
(a) Estimate the electron and hole concentrations at both temperatures.
(b) Use the exact equations to determine the concentrations at both temperatures.

WS2 Q6 (course question). At a temperature of 50 °C, for intrinsic silicon:
(a) Calculate the intrinsic carrier concentration n_i.
(b) Calculate the concentration of current carrying electrons (n) and holes (p).
(c) Calculate the resistivity of the silicon at this temperature.
(d) Calculate the resistance of a piece of silicon where the current is travelling a length of 5 µm over a cross-sectional area (perpendicular to the flow of current) of 3 µm × 3 µm.

(e) What is the difference (%) in resistance between what you calculated in (d) and the resistance of the silicon at room temperature (300 K)?

WS2 Q7 (course question). Find the resistivity and resistance according to the same constraints of Question 6 if the silicon is now doped with donors to a concentration of 5×10^{15} cm^{-3}.

PT Q1 (practice test, 5 marks). Circle the correct answer in the following multiple-choice questions (I to V).

I. When a PN junction is formed, why does a depletion region develop near the junction?

a) Electrons and holes recombine near the junction, leaving behind ionized dopants

b) The semiconductor becomes electrically insulating everywhere

c) All free electrons move from the N-region into the P-region permanently

d) The applied voltage removes all charge carriers from the semiconductor

II. The small-signal or dynamic resistance of a diode is related to:

a) The slope of the I–V characteristic at the operating point

b) The total length of the diode leads

c) The reverse breakdown voltage only

d) The DC supply resistance only

III. The main advantage of full-wave rectification over half-wave rectification is that full-wave rectification:

a) Uses only the negative half of the input waveform

b) Uses both half-cycles of the input waveform

c) Completely eliminates ripple without a capacitor

d) Requires no semiconductor devices

IV. Why is the base region of a BJT made relatively thin?

a) To allow most injected carriers to cross the base and reach the collector

b) To prevent carriers from reaching the collector

c) To increase the base resistance as much as possible

d) To eliminate the depletion region at the collector-base junction

V. Why are DC and AC analyses of a transistor amplifier often performed separately?

a) A transistor can operate with either DC or AC but never both

b) DC analysis establishes the operating point, while AC analysis examines small variations around that point

c) AC analysis determines the transistor's doping concentration
d) DC analysis is only needed when the transistor is damaged

New questions

N1. [5 marks] Circle the correct answer (I to V).

I. An n-type Si sample at 300 K has $N_D = 10^{16}\ \mathrm{cm}^{-3}$ ($n_i = 1.5 \times 10^{10}$). The hole concentration is:

a. 1.5×10^{10} b) 2.25×10^{4} c) 10^{16} d) 2.25×10^{20} cm⁻³

II. In a pn junction with no external bias:

a. No carriers cross the junction
b. The diffusion current equals the drift current, so the net current is zero
c. Only the drift current flows
d. The depletion region contains free electrons and holes

III. A junction has $N_A = 10^{17}$ and $N_D = 10^{15}$ cm⁻³. The depletion region:

a. Extends equally into both sides
b. Extends mostly into the p side
c. Extends mostly into the n side, with $x_n/x_p = 100$
d. Exists only on the p side

IV. A divider bias has R_1 from V_{CC} to the base node and R_2 from the base node to ground. The Thevenin resistance seen from the base is:

a. $R_1 + R_2$ b) $R_1 \parallel R_2$ c) R_2 only d) $R_1 R_2 / V_{CC}$

V. For an ideal op-amp with negative feedback, which pair is correct?

a. $v_+ = v_-$ and $i_+ = i_- = 0$
b. $v_o = 0$ and $i_+ = i_-$
c. $v_+ = 0$ and $v_- = v_o$
d. $i_+ = i_- = \infty$

N2. [6 marks] A p-type Si bar at 300 K has $N_A = 4 \times 10^{15}\ \text{cm}^{-3}$. Use $n_i = 1.5 \times 10^{10}$ cm⁻³, $\mu_n = 1350$ and $\mu_p = 500$ cm²/Vs. The bar is 2 mm long, with a rectangular cross-section of 0.5 mm × 0.2 mm. A voltage of 3 V is applied end to end.

a. Find the electron and hole concentrations. [1]
b. Find σ and ρ. Show the size of the minority-carrier term. [2]
c. Find the resistance and the current. [2]
d. Find the hole drift velocity. [1]

N3. [7 marks] (chained) A bias network has $V_{CC} = 15$ V, $R_1 = 68$ kΩ from V_{CC} to node B, and $R_2 = 12$ kΩ from node B to ground.

a. Find the Thevenin equivalent seen from node B. [3]
b. Node B is now connected to a branch made of an ideal 0.7 V drop in series with 101 kΩ to ground. (This is how a BJT base with $\beta = 100$ and $R_E = 1$ kΩ looks from the base: $(\beta + 1)R_E$.) Find the branch current I_B and the new voltage at B. [3]
c. A student ignores R_{Th} and uses $V_B = V_{Th}$. What percentage error does this give in I_B? [1]

Circuit for N3: the divider, with the 0.7 V + 101 kΩ branch connected in part (b).

N4. [7 marks] (chained) Node X is connected to +12 V through 20 kΩ, to −6 V through 30 kΩ, and to ground through 15 kΩ.

a. Find R_{Th} seen from X. [1]
b. Find V_{Th} by nodal analysis. Check your answer by superposition. [3]
c. A 5 kΩ load is connected from X to ground. Find V_X and the load current. [2]
d. Give the Norton equivalent. [1]

Circuit for N4: node X with three branches, and the 5 kΩ load added in part (c).

N5. [6 marks] A 12 V source connects through 2 kΩ to node 1. Node 1 has 4 kΩ to ground and 3 kΩ to node 2. Node 2 has 6 kΩ to ground. An independent 1 mA current source pushes current from ground into node 2.

a. Write the nodal equations and find V_1 and V_2. [4]
b. Find the power delivered by the current source, and state whether it supplies or absorbs power. [1]
c. Find the Thevenin resistance seen between node 2 and ground. [1]

Circuit for N5.

N6. [8 marks] (hard, chained) A lightly doped n-type Si sample has $N_D = 10^{13}\ \text{cm}^{-3}$. Use $B = 7.3 \times 10^{15}\ \text{cm}^{-3}\text{K}^{-3/2}$, $E_g = 1.12$ eV, $k = 8.62 \times 10^{-5}$ eV/K, $n_i(300\ \text{K}) = 1.5 \times 10^{10}$, $\mu_n = 1350$ and $\mu_p = 500$ cm²/Vs (assume constant).

a. Find n, p and σ at 300 K. [2]
b. Find n_i at 450 K. [1]
c. A student uses $n \approx N_D$ and $p = n_i^2/N_D$ at 450 K. Show why this is wrong. [1]
d. Find the correct n and p at 450 K. [2]
e. Find σ at 450 K, its ratio to the 300 K value, and V_T at 450 K. [2]

6. Quick recall check

1. At 27 °C, what is V_T?
2. Si at 300 K: what is n_i^2, and what is p in n-type material with $N_D = 10^{15}$?
3. When finding R_{Th}, what replaces a voltage source, and what replaces a current source?
4. Which side does the depletion region extend into more? Give the ratio formula.

5. For a divider bias (R_1 to V_{CC}, R_2 to ground), write V_{Th} and R_{Th}.
